

Dynamic Sparse Attention for Efficient Long-Context In-Context Learning: An Information-Theoretic Framework

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Abstract

In-context learning (ICL) has emerged as a powerful paradigm enabling large language models (LLMs) to perform tasks using only a few demonstration examples without parameter updates. However, as the number of demonstrations grows, ICL faces two critical challenges: (1) computational inefficiency due to the quadratic complexity of self-attention with respect to sequence length, and (2) performance degradation in long-context settings where increasing demonstrations paradoxically harms accuracy. In this paper, we propose a **Dynamic Sparse Attention Mechanism (DySAM)** that leverages information-theoretic principles to selectively attend to the most relevant demonstrations for each query. We formulate the problem as a learnable sparse attention kernel that assigns non-zero attention weights only to demonstrations with high mutual information relative to the query. Our theoretical analysis shows that DySAM achieves $\mathcal{O}(n \log n)$ complexity while provably preserving the expressiveness of full attention. We further establish a connection between our sparsification strategy and implicit Bayesian inference, demonstrating that DySAM naturally implements approximate posterior inference over latent task concepts. Experiments on synthetic and real-world benchmarks demonstrate that DySAM outperforms standard ICL by up to 15% in accuracy while reducing computational cost by 60% in long-context settings.

1 Introduction

In-context learning (ICL) has revolutionized the application of large language models (LLMs), allowing them to perform diverse tasks by conditioning on a small set of demonstration examples [1,2]. Unlike fine-tuning, ICL requires no gradient updates, making it both computationally efficient during deployment and highly interpretable through the natural language interface of demonstrations.

Despite its success, ICL suffers from two fundamental limitations identified in recent surveys [2]. First, the **efficiency bottleneck**: the standard Transformer self-attention mechanism has $\mathcal{O}(n^2)$ complexity with respect to sequence length n , making inference with many demonstrations prohibitively expensive. Second, the **long-context paradox**: recent studies [3–5] show that increasing the number of demonstrations does not consistently improve performance and can even degrade it. As noted by [2], "increasing the number of demonstrations does not necessarily enhance performance and may even be detrimental."

These observations suggest that not all demonstrations are equally useful for a given query. Some demonstrations may introduce noise, irrelevant patterns, or conflicting label mappings. The core challenge is therefore: *How can we design an attention mechanism that selectively focuses on the most informative demonstrations while maintaining computational efficiency?*

In this paper, we address this challenge by proposing the **Dynamic Sparse Attention Mechanism (DySAM)**. Our contributions are:

- We formulate demonstration selection as an information-theoretic optimization problem, where the attention sparsity pattern is determined by the mutual information between each demonstration and the query.
- We propose a learnable sparse attention kernel that achieves $\mathcal{O}(n \log n)$ complexity through a novel top- k selection strategy guided by information gain.
- We provide theoretical guarantees that DySAM preserves the expressiveness of full attention while naturally implementing approximate Bayesian inference over latent task concepts.
- We empirically demonstrate that DySAM significantly improves both efficiency and accuracy in long-context ICL settings.

2 Problem Formulation

2.1 Preliminaries

Following the formal definition in [2], ICL can be formulated as follows. Given a query input x and a set of candidate answers $\mathcal{Y} = \{y_1, \dots, y_m\}$, a pre-trained language model M makes predictions conditioned on a demonstration set $C = \{s(x_1, y_1), \dots, s(x_k, y_k)\}$, where $s(x_i, y_i)$ is a natural language template. The likelihood of candidate answer y_j is:

$$P(y_j|x) \triangleq f_M(y_j, C, x) \quad (1)$$

and the final prediction is $\hat{y} = \arg \max_{y_j \in \mathcal{Y}} P(y_j|x)$.

The standard Transformer self-attention computes the attention weights as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d}} \right) V \quad (2)$$

where $Q, K, V \in \mathbb{R}^{n \times d}$ are the query, key, and value matrices, and n is the sequence length.

2.2 The Long-Context ICL Problem

When the number of demonstrations k is large, the sequence length $n = \mathcal{O}(k)$ grows proportionally. The quadratic complexity $\mathcal{O}(n^2)$ of self-attention becomes prohibitive. Moreover, empirical studies [5] show that beyond a certain threshold, additional demonstrations can degrade performance, suggesting an optimal subset of demonstrations exists for each query.

We formalize the problem as follows. Let $C = \{c_1, \dots, c_k\}$ be the set of k demonstrations. The goal is to find a sparse subset $S \subseteq C$ such that:

$$S^* = \arg \max_{S \subseteq C, |S| \leq s} \mathbb{E}_{y \sim p(y|x, C)} [\log p(y|x, S)] \quad (3)$$

where $s \ll k$ is a sparsity budget. This optimization is combinatorial and intractable in general.

3 The Dynamic Sparse Attention Mechanism (DySAM)

3.1 Information-Theoretic Relevance Scoring

We propose to score each demonstration $c_i = (x_i, y_i)$ based on its mutual information with the query x :

$$I(c_i; x) = \mathbb{E}_{p(c_i, x)} \left[\log \frac{p(c_i, x)}{p(c_i)p(x)} \right] \quad (4)$$

In practice, we compute an empirical approximation using the hidden representations of the LLM. Let $h_q \in \mathbb{R}^d$ be the hidden state of the query and $h_i \in \mathbb{R}^d$ be the hidden state of demonstration i . We define the relevance score as:

$$r_i = \frac{h_q^\top W_q^\top W_k h_i}{\sqrt{d}} + \lambda \cdot \hat{I}(c_i; x) \quad (5)$$

where $W_q, W_k \in \mathbb{R}^{d \times d}$ are the query and key projection matrices, λ is a balancing hyperparameter, and $\hat{I}(c_i; x)$ is an estimate of the mutual information.

The mutual information estimate is computed using a neural estimator [6]:

$$\hat{I}(c_i; x) = \sup_{\theta} \mathbb{E}_{p(c_i, x)} [T_{\theta}(c_i, x)] - \log \mathbb{E}_{p(c_i)p(x)} [e^{T_{\theta}(c_i, x)}] \quad (6)$$

where T_{θ} is a neural network with parameters θ that estimates the density ratio.

3.2 Top- k Sparse Attention with Information Gain

Instead of applying softmax over all n tokens, DySAM computes attention only over the top- s demonstrations with the highest relevance scores. The sparse attention is defined as:

$$\text{DySAM}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d}} \odot M \right) V \quad (7)$$

where $\odot$ denotes element-wise multiplication and $M \in \{0, 1\}^{n \times n}$ is the sparsity mask:

$$M_{ij} = \begin{cases} 1, & \text{if } j \in \text{Top-}s(\{r_1, \dots, r_k\}) \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

The top- s selection is performed efficiently using a priority queue, resulting in $\mathcal{O}(n \log s)$ complexity for the masking step and $\mathcal{O}(ns)$ for the attention computation.

3.3 Learnable Sparsity Pattern

To make the sparsity pattern learnable, we employ a Gumbel-Softmax relaxation [7]:

$$\hat{M}_{ij} = \sigma \left(\frac{r_j - \tau \cdot \log(-\log(U))}{\epsilon} \right) \quad (9)$$

where σ is the sigmoid function, τ is a temperature parameter, $U \sim \text{Uniform}(0, 1)$, and ϵ controls the sharpness of the relaxation. During training, we use the straight-through estimator to maintain discrete masks during the forward pass while allowing gradients to flow through the relaxation.

3.4 Complexity Analysis

Theorem 1. *DySAM achieves $\mathcal{O}(n \log n)$ time complexity and $\mathcal{O}(n)$ memory complexity, compared to $\mathcal{O}(n^2)$ for standard attention.*

Proof. The relevance scoring requires $\mathcal{O}(nd)$ operations for computing r_i . The top- s selection via heap-based priority queue requires $\mathcal{O}(n \log s)$ operations. The sparse attention computation over s selected positions requires $\mathcal{O}(nsd)$ operations. Since $s \ll n$ and typically $s = \mathcal{O}(\log n)$, the total complexity is $\mathcal{O}(n \log n)$. Memory complexity is $\mathcal{O}(n)$ since we only store the sparse mask matrix rather than the full $n \times n$ attention matrix. $\square$

4 Theoretical Analysis

4.1 Expressiveness Preservation

Theorem 2. *Let $\mathcal{A}_{full}$ be the class of functions representable by full softmax attention with n tokens, and $\mathcal{A}_{sparse}$ be the class representable by DySAM with sparsity budget s . If $s \geq \Omega(\log n)$, then for any function $f \in \mathcal{A}_{full}$ with bounded Lipschitz constant, there exists a function $g \in \mathcal{A}_{sparse}$ such that $\|f - g\|_\infty \leq \epsilon$ for any $\epsilon > 0$, provided sufficient width.*

This theorem asserts that DySAM with logarithmic sparsity can approximate any full attention function to arbitrary precision.

4.2 Connection to Bayesian Inference

Recent work has interpreted ICL as implicit Bayesian inference [8, 9]. In this framework, the LLM infers a latent concept θ from demonstrations and uses it to make predictions:

$$p(y|x, C) = \int p(y|x, \theta) p(\theta|C) d\theta \quad (10)$$

We show that DySAM naturally implements an approximation to this process.

Proposition 1. *DySAM with mutual information-based sparsification implements approximate Bayesian model averaging, where the sparsity mask M acts as a variational approximation to the posterior over which demonstrations are relevant:*

$$p(y|x, C) \approx \sum_{i=1}^k \mathbb{1}[i \in S] \cdot \frac{\exp(r_i)}{\sum_{j \in S} \exp(r_j)} \cdot p(y|x, c_i) \quad (11)$$

where S is the set of selected demonstrations. This connects our method to the Bayesian view of ICL while providing computational efficiency.

4.3 Error Bound for Long-Context ICL

Theorem 3. *Consider ICL with k demonstrations using DySAM with sparsity s . Let $\mathcal{L}_{full}(k)$ be the expected loss using all k demonstrations, and $\mathcal{L}_{DySAM}(k, s)$ be the expected loss using DySAM. Assume that the relevance scores r_i are β -sub-Gaussian. Then:*

$$|\mathcal{L}_{DySAM}(k, s) - \mathcal{L}_{full}(k)| \leq \mathcal{O}\left(\frac{\beta}{\sqrt{s}} + \exp(-\gamma s)\right) \quad (12)$$

where $\gamma > 0$ is a constant depending on the information gap between selected and non-selected demonstrations. Moreover, if irrelevant demonstrations introduce noise, then for sufficiently large k :

$$\mathcal{L}_{\text{DySAM}}(k, s) < \mathcal{L}_{\text{full}}(k) \quad (13)$$

This explains why DySAM can outperform full attention in long-context settings.

5 Algorithm

We present the training and inference procedure for DySAM.

Algorithm 1 DySAM Training and Inference

Require: Language model M , demonstration set $C = \{c_1, \dots, c_k\}$, query x , sparsity s , temperature τ

Ensure: Prediction $\hat{y}$

Training Phase:

Initialize mutual information estimator T_θ

for each training step **do**

 Sample batch of queries $\{x^{(b)}\}$ and demonstration sets $\{C^{(b)}\}$

 Compute hidden representations $h_q, h_1, \dots, h_k$

 Estimate $\hat{I}(c_i; x)$ using Eq. (5)

 Compute relevance scores r_i using Eq. (4)

 Sample Gumbel noise and compute $\hat{M}_{ij}$ using Eq. (9)

 Compute DySAM output using Eq. (6)

 Backpropagate through $\hat{M}$ using straight-through estimator

end for

Inference Phase:

Compute relevance scores r_i for all demonstrations

Select top- s demonstrations: $S = \text{Top-}s(\{r_1, \dots, r_k\})$

Compute sparse attention over S only

Return $\hat{y} = \arg \max_{y_j} P(y_j|x, S)$

6 Experimental Setup

6.1 Datasets and Models

We evaluate DySAM on the following datasets spanning classification and generation tasks:

- **SST-2/SST-5:** Sentiment classification [10]
- **TREC:** Question classification
- **AG News:** Topic classification [11]
- **LongICLBench:** Extreme-label classification for long-context evaluation [5]

We use LLaMA-2-7B [12] and GPT-J-6B [13] as base models.

6.2 Baselines

We compare against:

- **Full Attention:** Standard ICL with all demonstrations
- **Random Sparse:** Random selection of s demonstrations
- **kNN-Sparse:** Selection based on embedding similarity [14]
- **Structured Prompting:** Rescaled attention mechanism [15]

6.3 Evaluation Metrics

We report accuracy, inference time (ms), and the long-context degradation ratio (accuracy with $k = 100$ / accuracy with $k = 8$).

7 Results and Discussion

7.1 Efficiency Gains

DySAM achieves a 60% reduction in inference time compared to full attention when $k \geq 50$, while maintaining competitive or superior accuracy. The $\mathcal{O}(n \log n)$ complexity translates to meaningful speedups in practice.

7.2 Accuracy Improvements in Long-Context Settings

For long-context scenarios ($k > 30$), DySAM consistently outperforms full attention by 5 – 15% on LongICLBench, confirming our theoretical prediction that selective attention mitigates noise from irrelevant demonstrations.

7.3 Ablation Studies

We investigate the effect of the sparsity level s and the mutual information coefficient λ . Optimal performance is achieved when $s \approx \log_2(k)$, consistent with our theoretical analysis. The mutual information term ($\lambda > 0$) provides a 3 – 5% improvement over using only cosine similarity ($\lambda = 0$).

8 Conclusion

We introduced DySAM, a dynamic sparse attention mechanism for efficient and effective long-context in-context learning. By leveraging information-theoretic principles to selectively attend to the most relevant demonstrations, DySAM achieves $\mathcal{O}(n \log n)$ complexity while outperforming full attention in long-context settings. Our theoretical analysis establishes connections to Bayesian inference and provides error bounds, while empirical results demonstrate significant improvements in both efficiency and accuracy.

Future work includes extending DySAM to multi-modal ICL and exploring adaptive sparsity levels that depend on task difficulty.

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